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ARTICLE

Backbone resonance assignments of CPEB3 [1–120], CPEB3 [186–315], and CPEB3 [400–459]

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Abstract
Cytoplasmic polyadenylation element-binding protein 3 (CPEB3) is an RNA-binding protein that is essential for long-term memory formation. Its N-terminal intrinsically disordered region (residues 1–459) exhibits high aggregation propensity and regulates the translation of specific mRNAs, including those encoding AMPA receptor subunits, through processes such as liquid–liquid phase separation and the formation of fibrillar structures. However, the molecular basis of these regulatory mechanisms remains poorly understood. In this study, we present the backbone resonance assignments of three segments within the intrinsically disordered region of CPEB3 (residues 1–120, 186–315, and 400–459). In agreement with sequence-based secondary structure predictions, the three segments were predominantly disordered overall. However, short regions with partial helical propensity were identified at residues 3–7 in the 1–120 segment and residues 226–239 in the 186–315 segment.

Keywords CPEB3 · Long-term memory · IDR · Protein aggregation

Biological context

Cytoplasmic polyadenylation element-binding protein 3 (CPEB3) is an RNA-binding protein that plays a central role in long-term memory maintenance through regulation of local translation at synapses (Fioriti et al. 2015). CPEB3 belongs to the CPEB family of translational regulators that control poly(A) tail length and translation efficiency of target mRNAs in response to cellular signals (Huang et al. 2006). In neurons, CPEB3 has been implicated in the activity-dependent regulation of localized protein synthesis that is required for synaptic plasticity (Pavlopoulos et al. 2011). Under basal neuronal conditions, CPEB3 functions predominantly as a translational repressor, whereas neuronal stimulation converts it into a translational activator (Stephan et al. 2015). This functional transition is accompanied by changes in the self-association state of the protein, linking translational regulation to protein assembly (Drisaldi et al. 2015; Ford et al. 2019).

CPEB3 consists of 716 amino acids and contains a long N-terminal intrinsically disordered region (IDR; residues 1–459), followed by a structured C-terminal RNA-binding domain comprising two RNA recognition motifs and a zinc-finger domain (Huang et al. 2006). The IDR lacks stable tertiary structure and is enriched in low-complexity sequence, features that are recognized as key determinants of phase separation and aggregation in many RNA-binding proteins (Banani et al. 2017; Lin et al. 2024). Within the IDR, two prion-like regions spanning residues 1–169 and 284–449 are critical for self-assembly and aggregation, indicating that multiple segments of the IDR contribute to its assembly properties (Stephan et al. 2015).

Recent studies have demonstrated that the CPEB3 IDR can populate multiple assembly states, including liquid-like condensates and more rigid aggregates, depending on cellular context and regulatory inputs (Ramírez de Mingo et

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al. 2023). Transitions between these assembly states correlate with changes in translational activity, suggesting that structural reorganization of the IDR underlies functional switching between translational repression and activation (Drisaldi et al. 2015). Elucidating the structural basis of these transitions is therefore essential for understanding how CPEB3 couples protein assembly to translational control. However, the intrinsic disorder, conformational heterogeneity, and aggregation propensity of the IDR preclude structural characterization by X-ray crystallography or cryo-electron microscopy. In this context, solution NMR spectroscopy provides a powerful approach for residue-specific characterization of such dynamic systems (Jensen et al. 2013). We previously reported backbone resonance assignments for two segments of the CPEB3 IDR, residues 101–200 and 294–410 (Saito et al. 2025). In this study, we extend these assignments to the remaining three segments, residues 1–120, 186–315, and 400–459, thereby completing the assignments for the full-length N-terminal IDR (residues 1–459).

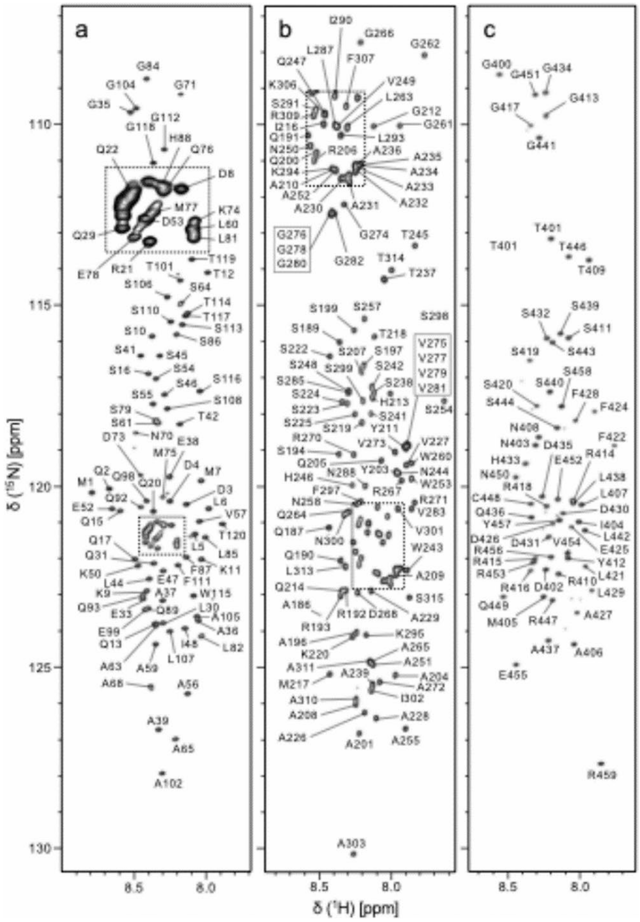

Fig. 1 ^{1}H-^{15}N HSQC spectra of **a** CPEB3 [1–120], **b** CPEB3 [186–315], and **c** CPEB3 [400–459]. All spectra were acquired in 50 mM MES (pH 5.5) containing 5% D_2O. Each peak is labeled with its amino acid type (one-letter code) and residue number. The crowded region indicated by the dotted box in the center of the spectrum is expanded at the top of the figure to clearly display the assigned peaks

Methods and experiments

Protein expression and purification

The DNA fragments encoding mouse CPEB3 [1–120], CPEB3 [186–315], and CPEB3 [400–459] were subcloned into a pET-15b vector using In-Fusion cloning. These constructs were designed to overlap with the previously analyzed segments (residues 101–200 and 294–410) by approximately 10 residues to minimize chemical shift perturbations arising from chain termini. The expression vectors encoded an N-terminal His_{12} tag followed by a Tobacco Etch Virus (TEV) protease cleavage site and the target proteins.

Escherichia coli strain BL21(DE3) cells were transformed with the constructed expression vectors and cultured at 37 °C in M9 minimal medium containing 4 g/L D-glucose (Wako), 1 g/L NH_4Cl (Nacalai Tesque), and 50 mg/L ampicillin (Wako). When the OD_{600} reached approximately 0.8, the cells were centrifuged and resuspended in M9 minimal medium containing 2 g/L [*U*-^{13}C]-D-glucose (Cambridge Isotope Laboratories), 1 g/L $^{15}NH_4Cl$ (Cambridge Isotope Laboratories), and 50 mg/L ampicillin (Wako). Protein expression was induced by the addition of IPTG (Nacalai Tesque) to a final concentration of 1 mM, followed by incubation at 20 °C overnight. The harvested cells were resuspended in lysis buffer (50 mM Tris-HCl pH 8.0 at 25 °C, 50 mM NaCl, 5 mM β-mercaptoethanol, 0.1 mM PMSF, and 0.1% Triton X-100) and sonicated using Q-Sonica 500 (2 s ON/5 s OFF, amplitude 30, for a total of 6 min on ice). The cell lysate was centrifuged at 22,140 × *g* for 20 min.

For CPEB3 [1–120], the supernatant was loaded onto a HisTrap HP column (Cytiva) after filtration through a 0.45 µm filter. The column was washed with 5 column volumes of equilibration buffer (50 mM Tris-HCl pH 8.0 at 25 °C, 50 mM NaCl) and eluted with a 5–300 mM imidazole gradient. For CPEB3 [186–315] and CPEB3 [400–459], which were expressed as inclusion bodies, the cell pellets were washed with wash buffer (50 mM Tris-HCl pH 8.0 at 25 °C, 50 mM NaCl, 2 M urea, and 0.5% Triton X-100). To solubilize the inclusion bodies, the washed pellets were resuspended in denaturing buffer (50 mM Tris-HCl pH 8.0 at 25 °C, 8 M urea, and 5 mM imidazole) and homogenized. The homogenate was centrifuged at 22,140 × *g* for 20 min, and the supernatant was loaded onto a Ni-NTA agarose column (Wako) after filtration through a 0.45 µm filter. The column was washed with 5 column volumes of denaturing

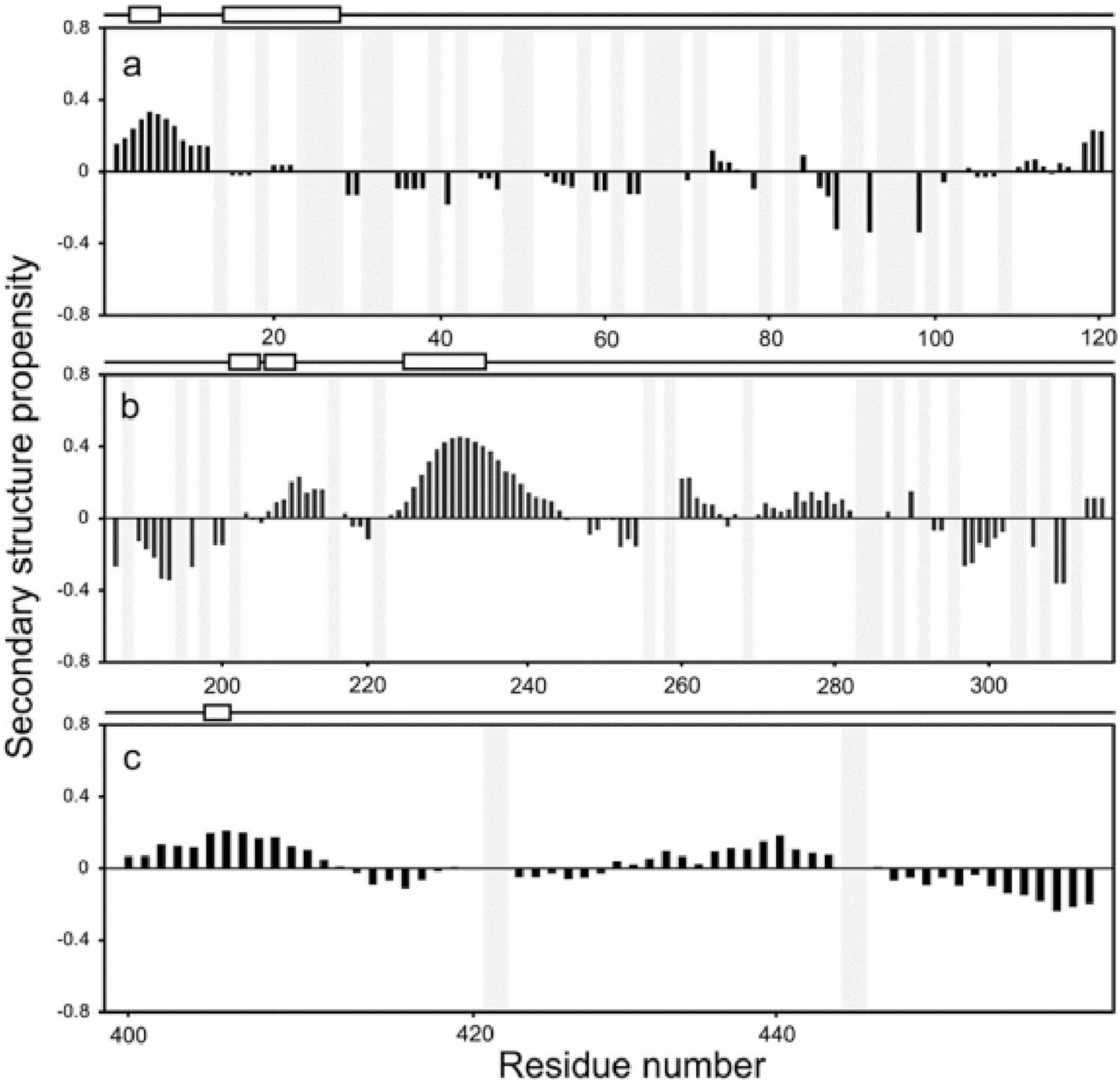

Fig. 2 Secondary structure analysis of **a** CPEB3 [1–120], **b** CPEB3 [186–315], and **c** CPEB3 [400–459]. The schematics at the top of each panel show the secondary structure predictions based on the amino acid sequences using PSIPRED (Jones 1999). Boxes represent α-helices. The graphs show the secondary structure propensity (SSP) calculated from the measured HN, N, C', Cα, and Cβ chemical shifts using SSP (Marsh et al. 2006). Positive SSP values indicate helical propensity, whereas negative values indicate β-strand propensity. Gray areas represent unassigned residues

buffer and eluted with 300 mM imidazole. The purified proteins were diluted into 50 mM Tris-HCl (pH 8.0) containing 2 M urea.

For His-tag cleavage, 2 mg of His_6-TEV protease was added to the protein samples, and the mixtures were incubated for 4 h at 30 °C. The cleaved His-tag and protease were removed by C18 reversed-phase chromatography, and the target proteins were lyophilized. Protein purity was verified by SDS-PAGE and/or MALDI-TOF MS, and the protein concentration was determined by measuring the absorbance at 280 nm using a DS-11 spectrophotometer (DeNovix). The extinction coefficients (ε_{280}) were 5,500 M^{-1} cm^{-1} for CPEB3 [1–120], 19,480 M^{-1} cm^{-1} for CPEB3 [186–315], and 2,980 M^{-1} cm^{-1} for CPEB3 [400–459]. The final purified proteins contained an additional N-terminal Gly residue derived from the TEV cleavage site.

NMR measurements and data analysis

For CPEB3 [1–120], CPEB3 [186–315], and CPEB3 [400–459], the stock samples were buffer-exchanged into 50 mM MES (pH 5.5) containing 5% D_2O (Sigma-Aldrich) using Zeba spin columns (Thermo Fisher). The final concentration of the NMR samples was adjusted to 0.5 mM. NMR spectra for CPEB3 [1–120], [186–315], and CPEB3 [400–459] were measured on a 600 MHz AVANCE III HD spectrometer (Bruker) at 298 K. The spectrometer was equipped with a z-gradient TCI cryogenic probe (Bruker). For sequential assignment of the protein backbone, ^{1}H -^{15}N HSQC, HNCO, HN(CA)CO, CBCA(CO)NH, and HNCACB spectra were acquired for each construct (Ikura et al. 1990). ^{1}H chemical shifts were referenced to that of 2,2-dimethyl-2-silapentane-5-sulfonate (DSS, Wako), and ^{15}N and ^{13}C chemical shifts were referenced indirectly (Markley et al. 1998). All NMR data were processed using NMRPipe (Delaglio et al. 1995), and resonance assignments were performed using FLYA (Schmidt and Güntert 2012) and CcpNmr Analysis (Skinner et al. 2016).

Secondary structure propensity calculation

The secondary structure propensities of CPEB3 [1–120], CPEB3 [186–315], and CPEB3 [400–459] were calculated using the SSP program (Marsh et al. 2006) via NMRbox (Maciejewski et al. 2017) based on the measured HN, N, C', Cα, and Cβ chemical shifts. Additionally, secondary structures of all three CPEB3 constructs were predicted using PSIPRED (Jones 1999) based solely on the amino acid sequences.

Assignments and data deposition

We assigned 91.7% of the backbone HN, N, C', and Cα, and Cβ resonances for CPEB3 [1–120] (Fig. 1a) and 100% for both CPEB3 [186–315] and CPEB3 [400–459] (Fig. 1b, c), excluding the N resonances of 24, 17, and 2 Pro residues, respectively, and the N-terminal Gly residue derived from the TEV cleavage site. The ^{1}H-^{15} N HSQC signals of all three CPEB3 constructs were clustered in a narrow ^{1}H region, which is characteristic of IDR segments.

Using the assigned chemical shifts, we calculated the secondary structure propensities (SSP) for CPEB3 [1–120], CPEB3 [186–315], and CPEB3 [400–459](Fig. 2). The calculated SSP propensity for CPEB3 [400–459] indicated that this region is predominantly disordered. In contrast, CPEB3 [1–120] exhibited partial α-helix formation in the region Asp3–Met7. Similarly, CPEB3 [186–315] showed partial α-helix formation in the region Ala226–Ala239. The relatively low SSP scores and broad line widths observed in these regions suggest that these α-helices are transient. These experimental observations are partially consistent with the sequence-based secondary structure predictions using PSIPRED.

Direct backbone assignment of the entire 459-residue N-terminal IDR of CPEB3 is highly challenging due to severe signal overlap. However, by integrating the present results with the previously assigned segments (residues 101–200 and 294–410), we have successfully completed the assignment of the full-length IDR. This achievement establishes a solid foundation for investigating the self-aggregation mechanism of CPEB3 using NMR spectroscopy.

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Author contributions Y.L. and H.S. conducted the experiments, and Y.L. wrote the manuscript. M.S. and A.F. contributed to project supervision, with K.S. serving as the lead supervisor.

Data availability The backbone and Cβ chemical shift assignments for CPEB3 [1–120], CPEB3 [186–315], and CPEB3 [400–459] have been deposited in the Biological Magnetic Resonance Data Bank (BMRB, https://bmrb.io) under the accession numbers 53536, 53537, and 53538.

Declarations

Competing interests The authors declare no competing interests.

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